What Makes a Game Last?

Early signals that help predict long-term player activity on Steam

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1. Motivation

We all play games. One of us was inspired by the creator of Stardew Valley and wants to build a game of their own, so we looked at the problem the way a small studio would: after a game launches, which early signs tell us whether people will still be playing months later? We do not have an answer yet. This report describes the data we built, a first look at it, and the models we plan to try.

We study games released from 2020 to 2024. That range gives many new releases, and every game in it has at least 12 months of follow-up data.

2. Problem

2.1. Goal

We want to predict how many players a game still has in the sixth month after launch, compared with its first month.

(1) **Y (what we predict):** M6/M1 ratio = average concurrent players in Month 6 ÷ average concurrent players in Month 1. A value of 0.50 means Month 6 has half the average concurrent players of Month 1; a value of 1.20 means 20% more. Month 1 is the first full calendar month after release; Month 6 is the sixth full calendar month.

(2) **X (what we predict from):** Three groups of early signals.

- Early performance: Month 1 average players, and the Month 1 to Month 2 decline, (M1-M2) ÷M1.
- Game characteristics: pricing model (free or paid), genre, multiplayer support, early access.
- Post-launch activity: number of official Steam announcements in the first 3 months, by type.

(3) **Metric:** Mean absolute error (MAE), the average gap between predicted and actual M6/M1 ratio.

Y is a ratio of concurrent players. It is not the share of individual players who come back. Our data only has monthly average concurrent players, so we cannot compute the day-1, day-7 and day-30 retention that mobile-game analytics usually track.

2.2. Potential Impact

The inputs are all known about three months after launch. If a model beats our baseline, a studio could see around Month 3 which games are on track to lose players, while it can still change course. We have not tested this yet, and so far we only have associations, not causes.

3. Related work

We read four studies. Two use Steam-wide data on many games, one studies the updates of a single game, and one follow individual players. The table shows what each did and how ours differs.

Study	Data	What they did and found	How ours differs
De Luisa et al. (2021)	8,000 Steam games released after	Predicted a game's second-month player count from its first-month count, price, languages, storage, release date and genres, with Bayesian models. The first-month count was by	They predict absolute counts at Month 2. We predict a relative measure

	2015; player counts from SteamDB	far the strongest predictor. Price had almost no effect. Sports, racing and simulation games tended to have higher counts, RPG and indie lower.	(M6 ÷ M1) and add early announcements.
Lin et al. (2018)	1,182 Early Access games on Steam	Studied how the Early Access model works on Steam. Developers update more often during Early Access, and the average review rating is higher then. 88% of Early Access games are indie.	They study Early Access games only. We compare Early Access with direct launch on the M6/M1 ratio.
Zhong and Xu (2022)	17,253 DOTA2 players; 25 updates (9 major, 7 minor, 9 bug-fix)	Compared three update types with fixed-effects and instrumental-variable models. Major, infrequent updates raised engagement (11% to 49%). Minor, frequent ones were not reliably helpful (−4.7% to 5.9%). Bug-fix updates made no clear difference. After COVID-19 began, updates had less effect.	One game. We count early announcements across thousands of games and cannot tell major from minor.
Hadiji et al. (2014)	About 20 million play sessions from 5 free-to-play games	Predicted which individual players stop coming back, with decision trees. In the more realistic setup, the time between sessions and the time since the last session mattered most.	They need per-player logs, which are usually not public. We study game-level player counts.

Among these four studies, none relates early signals to relative player activity at Month 6 across many Steam games. De Luisa et al. is the closest. Their authors suggest predicting relative change instead of absolute counts as future work, and our Y is a relative measure. We also add post-launch announcements as a signal. We did not search systematically, so we do not claim that no one has done this.

4. Data

4.1. What the data look like

Source	What it holds	Type	Size
SteamCharts	Monthly average concurrent players per game	Numeric	5,001 games, Months 1–12
Steam Storefront API and SteamSpy	Genre, free/paid flag and price, Early Access, single-player, multiplayer, co-op, tags, DLC count, achievements, languages, developer, publisher	Categorical, binary, counts. Genre has several labels per game.	6,015 games before filtering
Steam News API (official Community Announcements)	Title and date of each announcement	Text, turned into counts per game	272,786 announcements for 5,390 of the 6,015 games

- **Sample:** SteamDB lists about 65,500 games released in 2020-2024 (retrieved October 2026). We kept the 6,015 with the most estimated owners on SteamSpy (about 9%), using data collected by September 30, 2026. SteamSpy's owner counts are estimating not official Steam number. We then dropped 1,014 games with no player history, an incomplete first 12 months, or Month 1 average players of 0 or less. That leaves 5,001 games.
- **Final table:** 5,001 rows x 47 columns; one row is one game; the three sources are joined on Steam App ID. There are no duplicate App IDs. The table has 47 missing cells in total, all in descriptive metadata (publishers 23, genres 17, top tags 6, developer 1). Y and the announcement features have no missing values.
- **Announcements:** We collected 272,786 official Steam announcements for 5,390 of the 6,015 games. Of these, 92,016 (33.7%) were posted before release and are not used. 42,156 of the rest fall within the first 90 days, the window we use for post-launch activity. We label each announcement from its title with

keyword rules (for example, "patch" for game update, "% off" for promotion) as game update, new content, event, promotion, or other. One announcement can have several labels, so the shares add up to more than 100%. Of the 85,142 announcements in the first year, 40.5% match no category. The rest are game update (50.1%), promotion (5.6%), event (3.3%), and new content (2.5%). We checked two random samples of 100 titles and refined the rules after each. In the first, 88 needed no change, 11 missed a category, and 1 was wrongly labeled. In the second, 95 needed no change, and all 5 corrections were titles labeled "other" that belonged to a category, so some categories may be undercounted. We have not yet tested the final rules on a fresh sample. We plan to label a new random 100 for that.

4.2. The first look

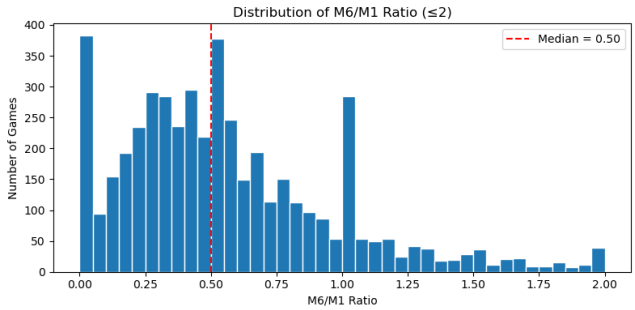


Figure 1: Distribution of M6/M1 Ratio
Median = 0.5

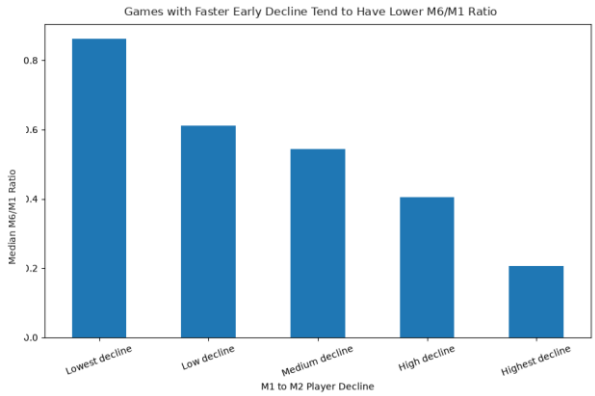


Figure 2: Median M6/M1 ratio in five equal groups, sorted by Month 1→2 decline

- **Distribution of Y:** Median 0.50, mean 0.99, middle half of games between 0.28 and 0.86. The maximum is 580. 4,755 games (95.1%) are at or below 2 and 246 are above. 335 games are exactly 0, 235 exactly 1.00 and 158 exactly 0.50.

Question	Test	Result (N = 5,001)
H1. Early performance	Spearman correlation with the M6/M1 ratio	• Month 1 players: ρ = −0.109. • Early decline: ρ = −0.485. • Both p < .001. Median M6/M1 ratio goes from about 0.86 (smallest-decline fifth) to about 0.21 (largest).
H2. Game characteristics	Mann–Whitney U, two-sided	• Paid (n = 4,274) vs. free (727): median 0.500 vs. 0.507, p = .602. • Single-player (3,448) vs. multiplayer (1,553): 0.500 vs. 0.508, p = .244. • Direct launch (4,583) vs. Early Access (418): 0.500 vs. 0.440, p = .063. • None is significant at .05. Genre medians range from 0.681 (Sports) to 0.449 (RPG), but genres overlap and we have not tested them.
H3. Post-launch activity	Spearman; Kruskal–Wallis across 5 groups of update announcements	• Update announcements: ρ = −0.052, p < .001. • New content −0.002, events −0.012. • promotions 0.000 (all p > .4). • Groups differ (H = 26.46, p < .001): median 0.545 for 1–2 update announcements, 0.418 for 11 or more.

- **How we read these:**

(1) With 5,001 games even very weak correlations can have small p-values, so statistical significance does not necessarily indicate a strong relationship. Labels for correlation strength very by field. Using a common rule of thumb, |ρ| below 0.2 is considered very weak, while 0.4 to 0.6 is considered moderate. Early decline (ρ = −0.485) shows a moderate negative association with the M6/M1 ratio, while initial popularity (ρ = −0.109) and update frequency (ρ = −0.052) show very weak negative associations.

(2) Early decline and the M6/M1 ratio share the Month 1 denominator. Specifically, M6/M1 = (M6/M2) X (M2/M1), while early decline = 1-M2/M1. This mathematical relationship may partly explain the observed correlation. We therefore treat early decline as an early warning signal available by Month 2, not as a cause of a lower M6/M1 ratio.

(3) In H3, "0 announcements" mean no game-update announcements were identified on Steam, not that the game received no actual updates.

(4) A p-values above .05 does not prove that no difference exists. The early access comparison was not statistically significant (p = .063), but we cannot rule out a difference between early access and direct-launch games.

(5) All findings describe associations, not causal effects.

4.3. Challenges we already see

- **Skewed Y:** The mean (0.99) is almost twice the median (0.50), and a few games are extreme.
- **Spikes and small counts:** The peaks at 0, 0.50 and 1.00, and common values such as 0.667 and 0.333, suggest some games have very few players in Month 1. When both monthly averages are small, the ratio can only take a few values. Figure 3 shows this at the left edge, where points fall on lines, and a few games average under one concurrent player in Month 1. We have not yet counted how many games have small Month 1 counts.
- **Approximate labels:** Genres overlap, and announcement types come from title keywords. About 40% of announcement titles match no category.

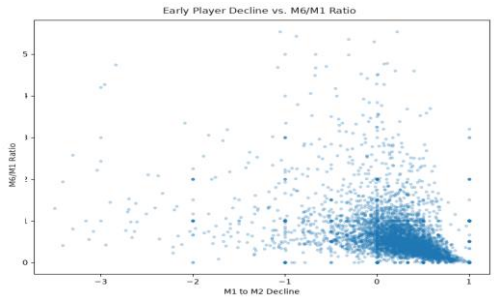


Figure 3: Month 1 average players (log scale) against the M6/M1 ratio, 99% of games. ρ = −0.109. Points line up because the ratio of two small counts takes only a few values.

4.4. Challenges we expect

- **Only popular games:** Results may not hold for less popular games. The 1,014 dropped games may have lost players faster than the rest, which could make the M6/M1 ratios in our sample appear higher than those of the broader game population. Owner counts run up to September 2026, so older games had more time to gather owners, and games that kept growing may rank higher.
- **Release timing:** Games launched in 2020–2021 may have had Month 1 during lockdowns. Zhong and Xu (2022) found that after COVID-19 began, players' intentions to play dropped, which weakened the effect of updates.
- **Cause and effect:** Studios may post more updates because players are leaving, so update counts could be a reaction, not a driver.
- **Many tests:** We ran about ten tests and did not adjust for that. Borderline results, such as p = 0.063, need

care.

5. Plan

- **Models:** Regularized linear regression, because it shows the direction and size of each factor, and a random forest, because it can capture patterns a straight line cannot. In H3 the median goes up for 1–2 announcements and then down.
- **Training:** We split the games 80% training and 20% testing. We tune settings on the training set only and use the test set once, at the end. We compare models by MAE on the test set.
- **Baseline:** Predict the training-set median M6/M1 ratio (about 0.50) for every game. It is the simplest prediction, using no features. The median is also the constant that gives the lowest MAE, and it is not pulled up by the extreme values that move the mean. A model is useful only if its test MAE is below the baseline's. We have not computed the baseline's test MAE yet.
- **Open decisions:**
  (1) Skewed Y: keep it as is and rely on MAE, or also try a log-transformed version.
  (2) Loss: both models minimize squared error by default. We may also try an absolute-error version.
  (3) Early decline shares numbers with Y. We may report results with and without it.
  (4) Testing on games outside our sample, for example ones released after 2024. Not decided.

6. References

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